

STT 465 BAYESIAN STATISTICAL METHODS

Fall 2026

Time and Location: Mon Wed Fri: 10:20 AM - 11:10 AM, Anthony Hall 1260

Instructor: Guanyu Hu, Office: Wells Hall C439 Email: huguanyu@msu.edu Office Hours: MW 11:30 AM - 1:00 PM (or by appointment)

Objects: This course introduces the fundamental principles and methods of Bayesian data analysis for undergraduate students with a background in probability and statistical inference. Bayesian statistics provides a coherent framework for learning from data by combining prior information with observed evidence to quantify uncertainty and make probabilistic decisions. Students will develop a strong understanding of Bayesian thinking, model construction, posterior inference, and predictive analysis through both theoretical concepts and practical applications.

The course begins with an introduction to Bayesian probability, prior and posterior distributions, likelihood functions, and Bayes' theorem. It then covers Bayesian estimation, credible intervals, hypothesis testing using Bayes factors, and Bayesian decision theory. Computational techniques, including conjugate analysis, Monte Carlo methods, Markov chain Monte Carlo (MCMC), Gibbs sampling, and Metropolis–Hastings algorithms, are introduced to enable inference for complex models. The course also explores Bayesian linear and generalized linear models, hierarchical Bayesian models, model comparison, model selection, and Bayesian prediction.

Course Materials:

- Introduction to Bayesian statistics and Bayesian thinking
- Foundations of probability and Bayes' theorem
- Prior, likelihood, posterior, and predictive distributions
- Conjugate Bayesian models
- Bayesian estimation and credible intervals
- Bayesian hypothesis testing and Bayes factors
- Bayesian decision theory
- Monte Carlo methods
- Markov chain Monte Carlo (MCMC)
- Bayesian linear regression
- Bayesian generalized linear models
- Hierarchical Bayesian models

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- Bayesian variable selection and shrinkage methods
 - Bayesian model comparison and model averaging
 - Posterior predictive inference and model checking
 - Missing data and Bayesian imputation
 - Bayesian causal inference
 - Bayesian machine learning
 - Bayesian data analysis using R
 - Applications to real-world problems in medicine, public health, business, engineering, environmental science, and sports analytics

Prerequisites: Knowledge of statistics and probability at the level STT 441& 442 is required throughout the course.

Exam and Homework Rules:

- Assignments: 5 homework assignments. (5% each) Homework assignments due approximately every 2 weeks. Late submissions will get a deduction of 15 % per late day.
- Midterm 1. (30%) Oct 23th
- Midterm 2. (30%) Dec 4th
- Projects: Students will be expected to have a final project report. (15%) Due Date: Dec 11th

Required Textbook: Peter D. Hoff (2009). A First Course in Bayesian Statistical Methods. Springer

Reference Books:

- Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., Rubin, D. B. Bayesian Data Analysis (3rd ed)

Midterm Policy:

Material policy: A numerical calculator is permitted. Students are permitted to use a cheat sheet during the exam. The cheat sheet must adhere to the following guidelines: Size: One standard letter-size (8.5" x 11") sheet of paper. Content: Both sides of the cheat sheet may be used. Format: The cheat sheet must be handwritten. No printed materials are allowed.

Regrade policy: Regrade requests will be reviewed by me. I will re-evaluate all questions using the grading rubric. The decision on the regrade request will be communicated to the student. The grade after re-evaluation may be higher, lower, or unchanged.

Final Projects:

The final project paper must be entirely double-spaced (abstract, main body, references, footnotes, and appendices), using at least 11 point type with at least 1-inch margins on all sides, and no longer than 35 pages (inclusive of all figures, tables, and bibliography).

The final project should do one or more of the following:

1. For real datasets, present analyses that are statistically innovative as well as scientifically and practically relevant.
2. Contribute substantially to a scientific field through the use of sound statistical methods.
3. Present new and useful data, such as a new life table for a segment of the population or a new social or economical indicator.
4. Using empirical tests, examine or illustrate for an important application the utility of a valuable statistical technique.
5. Evaluate the quality of important data sources.

Grading criteria will emphasize the strength of the manuscript in four areas:

1. Motivation by a relevant problem
2. Methodological development or investigation relevant to the motivating problem
3. Scientifically meaningful illustration of proposed methodology
4. Clarity and style of presentation

Grading Policy:

The default grade cutoffs are 90.0% for 4.0, 85.0% for 3.5, 80% for 3.0, 75.0% for 2.5, 70.0% for 2, 65.0% for 1.5, 60.0% for 1, and below 60% for 0. The actual grade cutoffs could be lower than this, if warranted by the distribution of scores.

Makeup Exams: Makeup exams will only be given for a legitimate reason (medical or family emergency or other exceptional circumstances) with verifiable proof. You must notify your instructor prior to missing an exam (whenever possible). Missing an exam without a valid excuse presented to your instructor will result in zero points for this exam.

Learning and Course Continuity: Students experiencing interruptions (e.g., illness, caregiving, or technical issues) should communicate with the instructor promptly. If you cannot progress for more than one week, email huguanyu@msu.edu for individualized accommodations. Similarly, the department will provide replacements should the instructor experience an extended absence.

Access, Opportunity, and Excellence: Michigan State University is committed to fostering a climate where **access**, **opportunity**, and **excellence** are central. I aim to ensure all students have the support needed to thrive, recognizing that our academic excellence is enriched by diverse perspectives. Students are expected to maintain professional courtesy to ensure an inclusive environment. If you encounter barriers to success or feel uncomfortable with any interaction, please contact me to discuss a resolution.

Attendance and Environment: Attendance is encouraged but not required. To maintain a respectful environment, please silence mobile devices. Laptops and tablets are permitted for note-taking and course-related tasks.

Communications: Announcements are made via D2L, email, and lectures. For the quickest response (within 24 hours on weekdays), email huguanyu@msu.edu.

Academic Integrity: The Department of Statistics and Probability adheres to MSU policies on academic honesty as specified in the *General Student Regulations* and *Spartan Life* handbook.

University Policies:

- **Religious Observance:** Notify the instructor at least two weeks prior to any conflict for accommodations.
- **Grief Absence:** See the university policy at <https://reg.msu.edu/ROInfo/Notices/GriefAbsence.aspx>.
- **Disabilities:** Contact the Resource Center for Persons with Disabilities (RCPD) at <https://www.rcpd.msu.edu/> to arrange accommodations.

Syllabus Changes: The instructor reserves the right to modify the syllabus as necessary, with changes announced in class.